Machine Learning - Undergraduate Level Assessment 18 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Compared to the variance of the Maximum Likelihood Estimate (MLE), the variance of the Maximum A Posteriori (MAP) estimate is ____________

   (a) higher

   (b) same

   (c) lower

   (d) it could be any of the above

2. Which of the following is the joint probability of H, U, P, and W described by the given Bayesian Network H -> U <- P <- W? [note: as the product of the conditional probabilities]

   (a) P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)

   (b) P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)

   (c) P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)

   (d) None of the above

3. Predicting the amount of rainfall in a region based on various cues is a _________ problem.

   (a) Supervised learning

   (b) Unsupervised learning

   (c) Clustering

   (d) None of the above

4. Statement 1| RELUs are not monotonic, but sigmoids are monotonic. Statement 2| Neural networks trained with gradient descent with high probability converge to the global optimum.

   (a) True, True

   (b) False, False

   (c) True, False

   (d) False, True

5. Consider the Bayesian network given below. How many independent parameters would we need if we made no assumptions about independence or conditional independence H -> U <- P <- W?

(a) 3
(b) 4
(c) 7
(d) 15

True / False (10 marks)

Answer True or False

6. True or False: The rank of the matrix A = [[1, 1, 1], [1, 1, 1], [1, 1, 1]] is 3.
7. True or False: Underfitting occurs when a model fails to capture the underlying patterns in the training data and performs poorly on new data.
8. True or False: P(A, B, | C) * P(C) accurately represents the joint probability of Boolean variables A, B, and C without independence assumptions.
9. True or False: Neural networks can use a mix of different activation functions to enhance learning and model complexity.
10. True or False: Lasso regression is more effective for feature selection compared to Ridge regression due to its ability to shrink some coefficients to zero.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to generate a two-dimensional array.
12. Write a function to find frequency count of list of lists.

End of Paper